

Sustainable Soilless Farming: The Role of Nutrient Media in Eco-Peat grown Hibiscus sabdariffa, Basella Alba

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ABSTRACT

With shrinking arable land, environmental degradation, and water scarcity threatening traditional agriculture, soilless farming emerges as a sustainable alternative. This study examines the impact of various nutrient media on the growth, yield, and nutritional composition of Sorrel (*Hibiscus sabdariffa*) and Malabar Spinach (*Basella alba*) grown in EcoPeat (cocopeat). Four nutrient media were tested: municipal water (control), hydroponic nutrient solution (N1), vermiwash (N2), and vegetable wash (N3). Growth indicators such as root and shoot length were assessed over four weeks, while nutritional parameters (moisture, ash, vitamins, and minerals) were analyzed using standardized laboratory techniques. Results indicate that hydroponic nutrient media (N1) consistently promoted superior growth and nutritional quality, whereas vermiwash (N2) and vegetable wash (N3) proved to be cost-effective, eco-friendly alternatives. This research highlights the potential of soilless farming in urban, resource-constrained environments and aligns with SDG 2 (Zero Hunger), SDG 12 (Responsible Consumption and Production), and SDG 15 (Life on Land). Future studies should explore optimized nutrient combinations to further advance sustainable urban farming practices.

Keywords: Soilless farming, Sustainable agriculture, Urban farming, Cocopeat, Eco-Peat, Nutrient media, Hydroponics, Hibiscus sabdariffa, Basella alba, Vermi-wash, Vegetable wash.

1. INTRODUCTION

Global population growth, urbanization, and land degradation have led to an urgent need for sustainable and space-efficient agricultural practices. Soilless farming has emerged as a modern solution capable of addressing constraints related to soil infertility, water scarcity, and limited arable land availability [1,5,11]. Soilless cultivation, particularly hydroponics, allows precise management of nutrients and water, offering a clean, controlled environment for crop production [4,13,15].

Cocopeat—a substrate derived from coconut husk—has gained popularity due to its high water-holding capacity, good aeration, and renewable nature [20]. However, cocopeat is nutrient-inert and must be supplemented with nutrient media. Nutrient delivery systems such as hydroponic solutions, vermiwash, and fermented vegetable wash differ in their

composition and impact on plant growth [2,6]. Comparable EcoPeat-based trials with other leafy vegetables have similarly shown that the choice of nutrient medium strongly influences both growth and nutritional outcomes [19]. In this study, we evaluate the performance of *Hibiscus sabdariffa* and *Basella alba*, two nutritionally rich leafy vegetables, under different nutrient treatments in a cocopeat-based

soilless cultivation system. The objective is to identify the most effective medium for promoting growth and yield in urban environments. This work extends earlier EcoPeat-based trials on *Amaranthus viridis* and *Spinaciaoleracea* [19] to two additional nutritionally important leafy vegetables, *Hibiscus sabdariffa* and *Basella alba*.

2. MATERIALS AND METHODS

2.1 Selection of Plants: Two green leafy vegetables were selected based on their nutritional and cultural significance: *Hibiscus sabdariffa* (Sorrel / Gogura), *Basella alba* (Malabar spinach/ Bachhali Kura)

2.2 Growing Substrate: Sterilized cocopeat was used in all pots as the soilless growing medium.

2.3 Nutrient Media: Four treatments were tested:

1. **Control (C):** Municipal water (pH 8–8.2, TDS < 100 ppm)
2. **Hydroponic Nutrient Solution (N1):** Prepared using a standard formula [6]
3. **Vermiwash (N2):** Liquid extract from 3-month vermicompost diluted 1:20 with water
4. **Vegetable Wash (N3):** Fermented vegetable waste solution (Bokashi method), diluted 1:20. These treatments followed the same protocol used in earlier comparative work on EcoPeat-grown leafy vegetables [19]

2.4 Nutrient Analysis

At the end of the fourth week, leaf samples of *Hibiscus sabdariffa* and *Basella alba* from each nutrient treatment were collected for following nutritional analysis to evaluate the influence of different nutrient media on the nutritional quality of the leafy vegetables.

Moisture content and **total ash** were determined following standard procedures described by Raghuramulu et al. (2003), as in previous EcoPeat cultivation studies [19]. **Vitamin C** was estimated using the 2,6-dichlorophenolindophenol (DCPIP) titration method (SAPS, 2010). **Calcium content** was determined by EDTA complexometric titration, **Iron** was estimated using Wong's colorimetric method (Raghuramulu et al., 2003).

2.5 Experimental Procedure

Seeds were soaked for 24 hours and sown in 5×5 inch thermoformed pots filled with cocopeat. The plants were maintained under ambient light and temperature with daily watering. Morphological data (root length, shoot length, and leaf count) were recorded weekly for four weeks. Four replicates per treatment were maintained.

3. RESULTS AND DISCUSSION

3.1.1 Growth Response of *Hibiscus sabdariffa* under Different Nutrient Media

Hibiscus sabdariffa plants were grown under four treatments: Control (C), Hydroponic Nutrient Solution (N1), Vermiwash (N2), and Vegetable Wash (N3). Over four weeks, root and shoot lengths were measured weekly to assess growth response. In Week 1, root growth was highest in N3, likely due to microbial activity and organic content [8], while shoot growth was unexpectedly highest in the control, possibly a stress response [18]. By Week 2, root growth continued to peak in N3, while shoot elongation was most significant in N2, likely due to plant growth regulators present in vermiwash [1]. In Week 3, the control showed increased root growth, while N1 recorded the tallest shoots, highlighting the steady effect of hydroponic nutrients [6]. By Week 4, the longest roots were observed in N3 (10.2 cm) and tallest shoots in N2 (19.6 cm), showing the effectiveness of organic nutrient media. Leaf count increased

Week →	1 st Week			2 nd Week			3 rd Week			4 th Week		
Media ↓	Leaf No	Root	Shoot	Leaf No	Root	Shoot	Leaf No	Root	Shoot	Leaf No	Root	Shoot
Control	4	3.2	11.4	6	4.9	13.5	7	8.9	12.1	8	9.2	15.6
N1	4	4.5	8.4	5	7.3	14.4	7	8.2	17.1	8	8	16.5
N2	4	4.5	10.8	6	8.3	16.6	6	8.3	16.5	9	8.5	19.6
N3	4	6.5	9.3	6	9.7	12.5	7	8.6	15.3	18	10.2	18.9

from 4 to 18 in N3 by Week 4, followed closely by N2. While N1 encouraged early floral bud sprouting, N2 and N3 were better for vegetative yield.

Overall, vegetable wash (N3) was most effective for root development, and vermiwash (N2) for shoot growth. These findings support previous studies on the benefits of organic nutrients in soilless systems [3, 7, 13]. A similar trend was reported in EcoPeat-grown *Amaranthus viridis*, where vegetable wash produced the highest leaf number and vegetative growth among all treatments [19]. Organic media also align with sustainable, low-cost farming practices ideal for urban agriculture [2, 11].

Table1: Weekly Comparative Analysis of Leaf Number, Root Length, and Shoot Length of *Hibiscus sabdariffa* Under Different Nutrient Media over four weeks

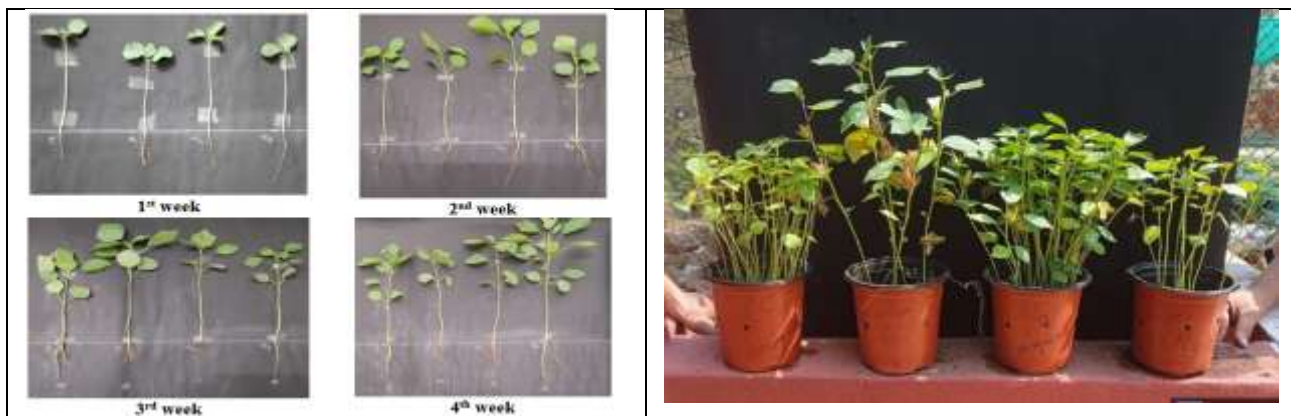


Figure 1: Growth of *Hibiscus sabdariffa*

Figure 2. Fully Grown Sorrel after 4 weeks

3.1.2 Nutritional Quality Assessment of *Hibiscus sabdariffa* under Different Nutrient Media

The nutritional composition of Sorrel varied significantly with nutrient media. Hydroponic treatment (N1) recorded the highest ash content and vitamin C concentration, indicating superior mineral accumulation and antioxidant synthesis under balanced nutrient supply [10,12]. Vermiwash (N2) enhanced iron accumulation, while Vegetable Wash (N3) resulted in higher moisture and calcium contents, producing more succulent foliage [20,16]. These results suggest that nutrient media influence specific nutritional attributes of Sorrel. Hydroponics favoured mineral and vitamin enrichment, Vermiwash improved micronutrient availability, and Vegetable Wash enhanced moisture retention and calcium uptake, demonstrating the potential of organic nutrient sources in soilless cultivation systems [20,12].

Table 2: Nutrient analysis of sorrel plants after 4th week

Media	Moisture content (%)	Ash (g)	Vitamin C (mg)	Calcium (mg)	Iron (mg)
Hydroponic media (N1)	79.2	19.0	26.25	107.2	3.0
Vermi wash (N2)	45.75	8.8	18.75	61.0	5.2
Vegetable wash (N3)	87.4	3.6	11.25	160.2	NR

3.2.1 Growth Response of *Basella alba* under Different Nutrient Media:

Malabar spinach (*Basella alba*) saplings were grown under four nutrient conditions: Control (C), Hydroponic Nutrient Solution (N1), Vermiwash (N2), and Vegetable Wash (N3). Weekly morphological data were collected over a four-week period to evaluate the effect of each treatment on root and shoot growth and leaf number. In **Week 1**, root growth was most prominent in the control, while shoot growth was highest in the vermiwash (N2) group. This suggests that early nutrient-limited conditions may trigger stronger initial root establishment, whereas vermiwash provided early hormonal stimulation for shoot development [1, 15]. By **Week 2**, both root and shoot growth peaked in the hydroponic group (N1), indicating rapid uptake and response to readily available mineral nutrients [6]. However, this growth trend did not sustain over time. In **Week 3**, root growth was highest in N2, while shoot elongation was most notable in vegetable wash (N3), which likely benefited from decomposed organic matter and microbial stimulation [3, 7]. By **Week 4**, root growth remained highest in N2 (7.9 cm), and shoot growth peaked in N3 (11.5 cm). A leaf count of 7 was recorded in both N2 and N3, reflecting robust vegetative performance.

Cumulatively, vermiwash (N2) showed the most consistent growth across all weeks, supporting earlier findings that vermiwash enhances root system development and shoot elongation due to its rich mix of enzymes and growth hormones [1, 4]. This parallels observations in *Spinacia oleracea*, where vermiwash consistently supported the most robust shoot development across the growth period [19]. Vegetable wash (N3) proved effective for shoot biomass, likely due to microbial enrichment and organic nutrient content [8, 13]. While hydroponic media (N1) showed strong early growth, it did not outperform the organic treatments by the end of the study. The control group (C) consistently lagged, affirming that cocopeat requires nutrient supplementation for sustained growth [1, 18].

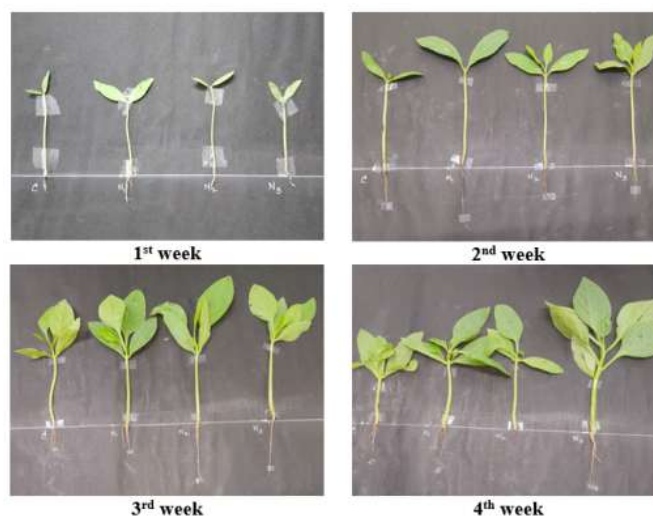


Figure 3. Growth of *Basella alba* under different Nutrient treatments over four weeks

Table 3: Weekly Comparative Analysis of Leaf Number, Root Length, and Shoot Length of *Basella alba* Under Different Nutrient Media

Week →	1stweek			2nd week			3rdweek			4thweek		
Media ↓	Leaf No	root	shoot	Leaf No	root	shoot	Leaf No	root	shoot	Leaf No	root	shoot
Control	2	1	7.1	3	4.2	8.5	5	3.3	9.8	9	5.4	6.1
N1	2	0.9	6.3	2	6.5	9.5	5	4.1	9	9	6	8.6
N2	2	0.7	7.6	4	3.2	9.3	7	7.8	9.5	7	7.9	9.5
N3	2	0.8	6.1	4	3.4	9.3	6	6.9	10.9	7	7.7	11.5



Figure 4. Fully grown Malabar Spinach under different nutrient treatments after 4 weeks

Leaf count reflected growth trends—starting at 4 leaves in Week 1 across all treatments. By Week 4, N3 rose sharply to 18 leaves, followed by N2 with 9, and N1 and C with 8 each. This highlights N3's effectiveness in promoting vegetative growth, likely due to steady nutrient release and organic matter decomposition [2, 8]. Overall, it was observed Vermiwash (N2) is the most effective medium for growing *Basella alba* in soilless systems, providing balanced and sustained root and shoot development. Vegetable wash (N3) also showed promise, particularly in shoot biomass, making both suitable for resource-efficient, organic urban farming.

3.2.2 Nutritional Quality Assessment of *Basella alba* under Different Nutrient Media

Nutrient composition of Malabar Spinach also varied among treatments. Hydroponic media (N1) produced the highest ash, vitamin C, and calcium contents, reflecting efficient mineral uptake under balanced nutrient supply [10,12,21]. Vermiwash (N2) recorded the highest iron concentration, while Vegetable Wash (N3) showed the greatest moisture content, contributing to improved leaf succulence and quality [9, 16]. The findings indicate that hydroponics enhances overall nutritional density, Vermiwash improves micronutrient enrichment, and Vegetable Wash supports moisture retention and leaf quality. Together, these results demonstrate that nutrient media affect not only plant growth but also the nutritional value of Malabar Spinach under soilless cultivation [19, 20].

Table 4: Nutrient analysis of Malabar Spinach plants after 4th week

Media	Moisture %	Ash (g)	Vitamin C (mg)	Calcium (mg)	Iron (mg)
Hydroponic media (N1)	79.7	16.7	39.45	208.6	NR
Vermiwash (N2)	80.0	1.56	NR	50	9.6

Vegetable wash (N3)	84.7	1.0	18.75	141	5.2
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(*NR Not reported)

4. CONCLUSION

This study evaluated the effects of hydroponic nutrient solution (N1), vermiwash (N2), and vegetable wash (N3) on the growth and nutritional quality of Hibiscus sabdariffa and Basella alba cultivated in cocopeat-based soilless systems. Vegetable Wash (N3) promoted superior root development, leaf production, and shoot growth, while Vermiwash (N2) supported consistent plant vigour and enhanced root development. Although hydroponic nutrient solution (N1) provided balanced growth and performed well during the early stages, the organic nutrient treatments (N2 and N3) proved equally effective or superior in sustaining overall plant development. The control treatment consistently recorded lower growth, emphasizing the importance of nutrient supplementation in cocopeat-based cultivation. These trends are consistent with earlier EcoPeat-based trials on Amaranthus viridis and Spinacia oleracea, which likewise found that nutrient-supplemented treatments outperformed the control across all growth parameters [19].

Nutrient analysis revealed that hydroponic treatment (N1) enhanced mineral accumulation, vitamin C, and calcium content, while Vermiwash (N2) improved iron enrichment. Vegetable Wash (N3) increased moisture content and leaf succulence, contributing to improved quality and palatability. These findings demonstrate that nutrient media influence both plant growth and nutritional composition. Overall, Vermiwash and Vegetable Wash emerged as sustainable, low-cost alternatives to conventional hydroponic solutions, supporting resource-efficient urban agriculture through organic waste recycling while improving crop productivity and nutritional quality. Taken together with earlier findings on Amaranthus and spinach [19], this study reinforces the broader potential of organic nutrient media across diverse leafy vegetable species in EcoPeat-based soilless systems.

5. ACKNOWLEDGEMENTS

The author sincerely thanks the Principal and the Director, Research & Development Cell, Government Degree College for Women, Begumpet, Hyderabad, for sanctioning the Minor Research Project under which this work was carried out and for their support in the successful completion of this study.

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